Project MO V1
3D Printing Specification and Quote Request

PURPOSE
Manufacture one complete V1 enclosure set from the supplied final 3MF files. Use every file at 100% scale in millimetres. Return printed parts only; do not create or modify client G-code.

General manufacturing requirements

Requirement	Specification	Requirement	Specification
Process	FDM / FFF	Material	PETG
Nozzle	0.4 mm	Layer height	0.20 mm
Perimeters	4	Infill	20% gyroid
Top / bottom	5 layers each	Scale	100%; no rescaling
Primary priority	Dimensional accuracy and functional fit	Color	Confirm with client before production

Part quantities and print setup

Part / final 3MF	Qty	Required orientation	Support / adhesion
Main Body	1	Rear exterior flat on bed; open front upward	Targeted build-plate support only where required beneath internal overhangs
FacePlate	1	Visible exterior flat on bed; internal bosses upward	None
Display Upper Retainer	1	Largest flat face on bed	None
Perfboard Retainers: UL, UR, LL, LR	1 each	Largest flat face on bed	None
D-pad Center Pivot	3	Flange on bed; stem vertical	None; 2-3 mm brim optional. Print one spare if practical
Left / Right Arm	1 each	Side-down; less-visible hand face toward bed	Organic/tree support from build plate
Left / Right Leg	1 each	Largest broad side on bed	Build-plate support and brim
D-pad, large circle, small circle, triangle buttons	1 each	Larger retaining flange on bed	None

FINAL FILE SET
Use the 16 files in **3MF_Final**. The set produces 18 physical pieces because the D-pad center pivot is printed three times. The older STL files and superseded 3MF exports are not production files.

Critical dimensions and interfaces

Feature	Final CAD dimension	Manufacturing note
Main enclosure envelope	Approx. 150 x 200 x 65 mm	Printer must accommodate the part with margin for adhesion and support
Main wall thickness	3.0 mm	Do not thin during mesh repair
Material below button pockets	1.5 mm	Preserve pocket depth and visible front surface
FacePlate perimeter clearance	0.50 mm per side	Centered clearance; do not scale either mating part
Display lower support clearance	0.30 mm	Preserve revised gravity-side support position
Display rear retainer clearance	0.20 mm	Preserve stepped rear support and 2.0 mm capture lip
Arm peg / socket	6.0 x 6.0 mm / 6.4 x 6.4 mm	0.20 mm nominal clearance per side; socket passes through 3 mm wall
Leg peg / socket	5.0 x 5.0 mm / 5.6 x 5.6 mm	0.30 mm nominal clearance per side; socket passes through 3 mm wall
D-pad center pivot	7.5 mm nominal total; 2 mm stem; 5 x 1 mm flange	Print three copies; keep flange flat
Grille / vent minimum spacing	1.5 mm	Avoid compensation that closes openings

Final switch interface

Feature	Dimension / status	Requirement
Panel cutout	Diameter 20.00 mm	Do not enlarge unless a physical switch fit test requires it
Selected switch body	Approx. diameter 19.6 mm	Seller-confirmed body dimension; verify purchased part
Upper flange	Diameter 22.9 mm	Seller-confirmed flange dimension
Internal clearance envelope	Diameter 19.6 x 30 mm, provisional	Physical switch depth and terminal clearance remain purchase-validation items
Nearby material	Approx. 4.5 mm minimum to lower flange region	Preserve the revised 1 mm upward clearance adjustment

Fastener interfaces

Application	Screw	Clearance / pilot information
FacePlate to body	M3 x 10 mm self-tapping / thread-forming	FacePlate clearance: 3.4 mm; body pilot: 2.6 mm. Confirm depth before production
Display Upper Retainer	M3 x 12 mm self-tapping / thread-forming	Retainer clearance: 3.4 mm; body pilot: 2.6 mm. Confirm at least 6 mm usable pilot depth
Button perfboard retainers	M2 x 8 mm self-tapping / thread-forming	Retainer clearance: 2.2 mm; FacePlate support-boss pilot: 1.6 mm x 4.5 mm deep; approx. 4.4 mm thread engagement

PERFBOARD RETAINER RELEASE DIMENSIONS
Final board overlap: **2.929 mm**. Side fillet: **R2.00 mm**. Minimum material observed around the M2 hole: approximately **2.76-2.90 mm**. The overlap was checked and does not obstruct required solder pads.

File manifest and verification

Final file group	Files	Printed pieces	Status
Enclosure	Main Body; FacePlate	2	Re-imported and sliced
Internal retainers	Display Upper Retainer; four Perfboard Retainers	5	Re-imported and sliced
Controls	D-pad; large circle; small circle; triangle	4	Re-imported and sliced
D-pad pivots	One 3MF, printed three times	3	Re-imported and sliced
Limbs	Left / right arms; left / right legs	4	Previously verified; unchanged
TOTAL	16 unique 3MF files	18	Release candidate

Quality and delivery requirements

CHECK Confirm the main body fits the selected machine before accepting the order.

CHECK Use only the files inside 3MF_Final and keep every object at 100% scale in millimetres.

CHECK Do not globally scale, smooth, hollow, decimate or repair geometry in a way that changes functional dimensions.

CHECK Keep supports away from button openings, limb sockets, screw pilots, display guides and visible exterior faces wherever possible.

CHECK Remove support and brim without enlarging mating features; leave cosmetic finishing to the client unless separately quoted.

CHECK Quote all parts in PETG and identify any part recommended for a separate build plate or different orientation.

CHECK Do not use files stored in Legacy_STL or Old_Versions for production.

PRE-PRODUCTION VALIDATION
The Nextion NX8048P050-011R envelope, revised display supports, enclosure meshes and retainer meshes were checked in CAD and PrusaSlicer. Before full production, physically verify the purchased power switch depth and terminals, display rear connector and cable route, ESP32 variant, amplifier, speaker and perfboard dimensions. Print small fit coupons for the 20 mm switch cutout, screw pilots, button clearances and limb peg/socket interfaces when practical.

Release package

Item	Location / description
Final meshes	3MF_Final - 16 unique 3MF files
Printing manifest	PRINT_MANIFEST.md - quantities and verification status
Editable CAD archive	Fusion_Source/Project_MO_V1_Manufacturing_Release.f3z
Legacy files	Legacy_STL and Old_Versions - backup only; not for production